What Lean Cannot Say: A Machine-Checked Analysis of Wittgenstein's *Tractatus*

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Abstract

We present a machine-checked reconstruction of the semantic core of *Tractatus Logico-Philosophicus* in the Lean proof assistant. The formalization models objects, states of affairs, worlds, and propositions, and proves truth-functional compositionality, functional completeness, and bivalence.

Beyond encoding, the system is parameterized over world models, allowing a precise separation between (i) structural invariants of the propositional syntax, (ii) model-dependent assumptions such as the independence of atomic facts, and (iii) formal limits of expressibility. We show that truth-functional compositionality holds in all models, while independence characterizes a specific "free" model and fails in constrained models.

To analyze the relation between object language and metalanguage, we introduce an `expresses` relation linking propositions to meta-level predicates. Using this, we prove that any proposition expressing a world-independent property must be a tautology or contradiction. Consequently, nontrivial propositions cannot express world-independent truths.

We further distinguish three notions of equivalence—syntactic identity, logical-form equivalence up to permutation, and semantic equivalence—and prove their strict separation.

These results yield a formal decomposition of the *Tractatus* into invariants, assumptions, and limits, and provide a precise characterization of the saying/showing distinction as a theorem about expressibility in truth-functional languages. The formalization comprises 1,517 lines of Lean 4, proves 50 formally verified results with 0 `sorry` obligations and 1 deliberate axiom (`axiom silence : True`, encoding TLP 7).[1]

Keywords: Wittgenstein, *Tractatus Logico-Philosophicus*, proof assistant, Lean 4, formal verification, truth-functional semantics, saying/showing distinction, expressibility

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1 Introduction

Wittgenstein's *Tractatus Logico-Philosophicus* [1] proposes that the world consists of atomic facts (Sachverhalte), that propositions are truth-functions of elementary propositions, and that the limits of language are the limits of the world. The text is terse, aphoristic, and deliberately resistant to formalization—yet its semantic core is a precise theory of truth-functional composition that invites formal reconstruction.

Several authors have pursued this invitation. Lokhorst [3] provides a set-theoretic reconstruction covering ontology, semantics, and propositional attitudes; Fogelin [4] initiates the debate over expressive completeness that Geach [8] and Soames [9] develop; Miller [10] develops a Tractarian semantics for predicate logic, responding in part to Fogelin and Carruthers [5]; Weiss [7] reconstructs the logical system with Π^1_1-completeness results; Stokhof [6] traces the *Tractatus*'s influence on formal semantics. These reconstructions clarify the *Tractatus* as a semantic system but are carried out on paper, without machine verification. Separately, the

[1] The formalization is publicly available at `https://github.com/rjwalters/tractatus`, with a browsable proof gallery at `https://leangenius.org/proof/tractatus-ontology`. It builds with Lean 4 (toolchain `leanprover/lean4:v4.17.0`), depends on Mathlib (`import Mathlib.Tactic`), and compiles via `lake build`. See Appendix A for full instructions.

proof-assistant community has formalized philosophical arguments [11, 12] and even complete metaphysical systems [13], but a Tractarian formalization has not appeared.

This paper contributes a machine-checked reconstruction in Lean 4 [15] that goes beyond encoding to yield new analytical results. Our central claim is:

> *The saying/showing distinction of the* Tractatus *can be reformulated as a theorem: nontrivial propositions cannot express world-independent truths.*

This is not a paraphrase of Wittgenstein but a provable consequence of the typed structure of the formalization. The key device is an `expresses` relation (Definition 5.1) that bridges the object language (propositions of type `Proposition S`) and the metalanguage (terms of type `Prop`), making the object/meta boundary visible at the type level.

More broadly, by parameterizing the formalization over a general `WorldModel` structure, we show that the claims of the *Tractatus* decompose into three classes:

1. **Structural invariants** that hold in every world model—compositionality, bivalence, De Morgan duality.
2. **Model-dependent assumptions** that hold in the unconstrained "free" model but fail in constrained models—principally, the independence of atomic facts.
3. **Formal limits** on expressibility—the collapse of world-independent content to tautology or contradiction, and the strict separation of syntactic, logical-form, and semantic equivalence.

The specific contributions of this paper are:

(1) A parameterized world-model abstraction that cleanly separates structural invariants from modeling assumptions, enabling precise identification of which Tractarian claims are theorems of the syntax and which are properties of a particular world model (Section 2).
(2) An `expresses` relation that makes the object/meta boundary a typed distinction, yielding the expressibility collapse theorem (Theorem 5.2): any proposition expressing a world-independent property is a tautology or contradiction (Section 5).
(3) An equivalence hierarchy $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$ with strict separation witnesses, providing a formal candidate for Wittgenstein's "logical form" (Section 5.3).
(4) A three-way decomposition of all 50 formalized results into invariants, assumptions, and limits (Table 1, Appendix B).
(5) A first-order extension using higher-order abstract syntax, proving compositionality lifts to quantified propositions (Section 5.4).

The formalization comprises 1,517 lines of Lean 4 across two files.[2] All proofs compile with 0 `sorry` obligations. The sole axiom is `axiom silence : True` (TLP 7), a deliberate boundary marker discussed in Section 7.2.

The paper is organized as follows. Section 2 presents the formalization: types, evaluation, and the world-model abstraction. Section 3 establishes the structural invariants. Section 4 analyzes the model-dependent assumptions and exhibits countermodels. Section 5 proves the expressibility results and the equivalence hierarchy. Section 6 discusses related work. Section 7 reflects on the philosophical implications and limitations.

2 The Formalization

We formalize the ontological skeleton of the *Tractatus* in Lean 4. The design choices are deliberate modeling decisions, not exegetical claims about Wittgenstein's intent. We state them explicitly so that the reader can assess the reconstruction's scope and limitations.

Remark 2.1 (Choice of proof assistant). We use Lean 4 [15] rather than Coq, Isabelle, or Agda for three reasons: (i) its dependent type theory allows definitions and proofs to coexist naturally, which suits a formalization that mixes data definitions with metatheorems; (ii) the Mathlib library provides automation (`simp`, `tauto`, `push_neg`) that keeps proofs concise; and (iii) Lean's tactic mode allows readable proof scripts that can be presented in a paper without excessive notation overhead.

[2] `TractatusOntology.lean` (1,229 lines, 41 theorem/lemma declarations) and `TractatusQuantifiers.lean` (288 lines, 9 theorem declarations), plus 1 deliberate axiom. Appendix B lists every declaration.

Remark 2.2 (Classical logic). The formalization assumes classical logic throughout, via `Classical.em` and related principles imported from Mathlib. This is a deliberate choice matching the *Tractatus*'s bivalent semantics: every proposition is either true or false in every world (TLP 4.023). Theorems depending essentially on classical reasoning include bivalence, excluded middle as a tautology, double negation, the expressibility collapse theorem, and contingent-proposition variation. A constructive formalization would require restricting the world model or weakening several results.

2.1 Objects and States of Affairs

TLP 2.02: "The object is simple." Objects are the substance of the world (TLP 2.021). We model them as an abstract type `TractObject` with no internal structure, capturing Wittgenstein's insistence on simplicity.

A state of affairs (Sachverhalt) is a possible combination of objects (TLP 2.01). Rather than encoding the combinatorial structure of objects into states of affairs, we index states of affairs by an abstract type S:

```
variable (TractObject : Type)
variable (Sachverhalt : Type)
```

This choice leaves the internal structure of states of affairs unspecified—an interpretive decision that captures independence while remaining agnostic about combinatorial form (cf. TLP 2.0141).

2.2 Worlds

TLP 1: "The world is everything that is the case." A world is determined by which states of affairs obtain. We model this as a predicate:

```
def World := Sachverhalt →Prop
```

Every function $\mathsf{S} \to \mathsf{Prop}$ counts as a world. This gives the full Boolean cube of truth-value assignments, making the independence of atomic facts (TLP 2.061) hold by construction. We return to this modeling choice and its consequences in Section 4.

2.3 Propositions

TLP 5: "The proposition is a truth-function of elementary propositions." We define propositions as an inductive type built from elementary propositions, negation, and conjunction:

```
inductive Proposition (S : Type) where
  | elementary : S →Proposition S
  | neg        : Proposition S →Proposition S
  | conj       : Proposition S →Proposition S →Proposition S
```

Disjunction, implication, biconditional, and NAND are defined as derived operations (TLP 5.101, 5.5), confirming that $\{\neg, \wedge\}$ is an adequate basis.

2.4 Semantic Evaluation

The core semantic function interprets propositions against a world:

```
def Proposition.eval (p : Proposition S) (w : World S) : Prop :=
  match p with
  | .elementary s => w s
  | .neg q       => ¬(q.eval w)
  | .conj q r    => q.eval w ∧r.eval w
```

An elementary proposition `elementary` s is true in world w exactly when $w\ s$ holds. Negation and conjunction lift to their classical counterparts.[3]

2.5 Tautology, Contradiction, Nontriviality

A proposition is a *tautology* if it holds in every world, a *contradiction* if it holds in none, and *nontrivial* if it holds in some world and fails in another:

```
def IsTautology (p : Proposition S) : Prop :=
  ∀ w : World S, p.eval w
def IsContradiction (p : Proposition S) : Prop :=
  ∀ w : World S, ¬(p.eval w)
def Nontrivial (p : Proposition S) : Prop :=
  ∃ w₁ w₂ : World S, p.eval w₁ ∧¬p.eval w₂
```

We prove that nontriviality is equivalent to being neither a tautology nor a contradiction.[4]

2.6 General World Models

The definitions above hardwire the world model: every function $\mathsf{S} \to \mathsf{Prop}$ is a world. In many applications—physics, epistemic logic, constrained databases—not every assignment is legitimate. We therefore introduce a `WorldModel` structure that parameterizes the semantics:

```
structure WorldModel (S : Type) where
  W       : Type
  holds   : W →S →Prop
  nonempty : Nonempty W
```

The *free model* `freeModel` recovers the original semantics: $W = \mathsf{S} \to \mathsf{Prop}$ and `holds` is function application. We define a general evaluator `evalM` and prove that `evalM (freeModel S)` agrees with `Proposition.eval` (Theorem `evalM_free_eq_eval`).

This abstraction is the key structural device of the formalization. It allows us to state precisely which results depend on the choice of world model and which do not.

3 Structural Invariants

The following results hold in *every* world model—they are consequences of the inductive structure of propositions and classical logic alone, with no assumptions about which worlds exist.

3.1 Truth-Functional Compositionality

Theorem 3.1 (Compositionality, model-independent)**.** *For any world model M, any proposition p, and any two worlds $w_1, w_2 \in M.W$: if $M.holds\ w_1\ s \leftrightarrow M.holds\ w_2\ s$ for every state of affairs s, then* $\mathsf{eval}_M\ p\ w_1 \leftrightarrow \mathsf{eval}_M\ p\ w_2$.

The proof is by structural induction on propositions. The elementary case follows from the hypothesis; negation and conjunction lift via `Iff.not` and `Iff.and`.

```
theorem truth_functional_compositionality_gen (p : Proposition S)
    (w₁ w₂ : M.W)
    (h : ∀s : S, M.holds w₁ s ↔M.holds w₂ s) :
    evalM M p w₁ ↔evalM M p w₂ := by
```

[3]The formalization also provides a `Bool`-valued evaluator `evalBool` for decidable computation and proves its correctness against `eval` (Theorem `evalBool_correct`). This engineering artifact enables `#eval` demonstrations but is not philosophically load-bearing.

[4]Theorem `nontrivial_iff_not_taut_not_contra` in `TractatusOntology.lean`.

```
induction p with
| elementary s => exact h s
| neg q ih    => simp only [evalM]; exact ih.not
| conj q r ihq ihr => simp only [evalM]; exact ihq.and ihr
```

This establishes TLP 5 as a structural invariant: the truth-value of a compound proposition is determined by its elementary constituents regardless of which worlds the model admits.

3.2 Bivalence and Classical Tautologies

Elementary bivalence (TLP 4.023) is an instance of classical excluded middle:

Theorem 3.2 (Bivalence)**.** *For any state of affairs s and world w: $w\ s \vee \neg\ w\ s$.*

The standard tautologies follow: excluded middle ($p \vee \neg p$), double negation ($\neg\neg p \leftrightarrow p$), De Morgan's laws, and modus ponens. We verify each as a formal theorem. All of these results depend on `Classical.em` (see Remark 2.2).

3.3 NAND Functional Completeness

TLP 5.5 anticipates the Sheffer stroke. We prove that negation and conjunction are both expressible via NAND:

Theorem 3.3 (NAND completeness)**.** *For any propositions p, q and world w:*

1. $\mathsf{nand}(p, p).\mathsf{eval}\ w \leftrightarrow (\neg p).\mathsf{eval}\ w$
2. $(\neg\ \mathsf{nand}(p, q)).\mathsf{eval}\ w \leftrightarrow (p \wedge q).\mathsf{eval}\ w$

Since $\{\neg, \wedge\}$ generates all truth-functions and NAND generates $\{\neg, \wedge\}$, NAND alone is functionally complete over the set of truth-functions expressible by the proposition type—confirming TLP 5.5.

4 Model-Dependent Assumptions

Not every claim in the *Tractatus* is a structural invariant. The independence of atomic facts—perhaps the most famous ontological thesis of the text—is a property of a specific world model.

4.1 Independence in the Free Model

We formalize independence as a typeclass:

```
class IndependentWorlds (S : Type) where
  realizable : ∀assignment : S →Prop,
    ∃ w : World S, ∀s, w s ↔assignment s
```

The free model trivially satisfies this: every assignment *is* a world.

```
instance : IndependentWorlds S :=
  ⟨fun a => ⟨a, fun _ => Iff.rfl⟩⟩
```

Independence is thus not a theorem about logic but a consequence of the modeling choice $\mathsf{World}\ \mathsf{S} = \mathsf{S} \to \mathsf{Prop}$. This distinction—between what is proved and what is assumed by the encoding—is precisely the kind of clarification that a typed formalization forces.

4.2 Failure in Constrained Models

To show that independence is genuinely contingent on the model, we exhibit two countermodels.

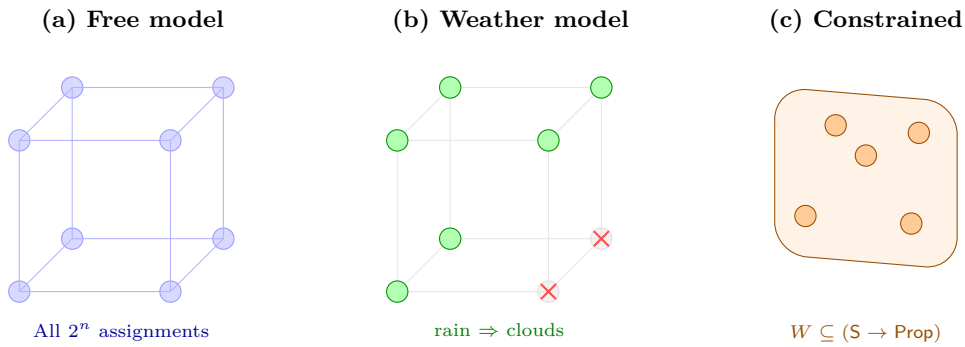

Figure 1: World-model architecture. (a) The free model admits all 2^n truth-value assignments as worlds. (b) The weather model excludes assignments violating rain ⇒ clouds (crossed vertices). (c) An abstract constrained model selects an arbitrary subset of the Boolean cube. Compositionality holds in all three; independence holds only in (a).

Abstract countermodel. Consider two states of affairs a, b with $a \neq b$, where b must co-occur with a. Define `ConstrainedWorld` $\mathsf{S}\ a\ b = \{w : \mathsf{S} \to \mathsf{Prop} \mid w\ a \to w\ b\}$. Then:

Theorem 4.1 (Independence fails in constrained models)**.** *If $a \neq b$, then there is no constrained world realizing the assignment $\{a \mapsto \top, b \mapsto \bot\}$.*

```
theorem constrained_independence_fails (a b : S) (hab : a ≠b) :
    ¬ ∀ assignment : S →Prop,
      ∃ cw : ConstrainedWorld S a b,
        ∀ s, cw.val s ↔assignment s := by
  intro h
  let bad : S →Prop := fun s => s = a
  obtain ⟨⟨w, hw⟩, hmatch⟩ := h bad
  have ha : w a := (hmatch a).mpr rfl
  have hb : w b := hw ha
  exact hab ((hmatch b).mp hb)
```

Weather model. A more concrete example: three states of affairs (rain, clouds, snow) with the physical constraint *rain implies clouds*. The weather model restricts worlds to those satisfying this constraint.

Theorem 4.2 (Weather model)**.** *In the weather model, there is no world where rain holds but clouds does not.*

These results make explicit a point that is often asserted informally: TLP 2.061 ("Atomic facts are independent of one another") is not a logical law but a constraint on the space of possible worlds. Our formalization turns this observation into a proof.

Notably, Wittgenstein himself later retreated from the independence thesis. In "Some Remarks on Logical Form" [2], he acknowledged that the color-exclusion problem—the mutual incompatibility of determinate color predicates (e.g., a point cannot be both red and blue)—demonstrates that certain atomic facts are not logically independent. Our constrained models formalize precisely this kind of failure: the weather model's constraint that rain implies clouds is structurally analogous to the color-exclusion problem. In both cases, physical or conceptual constraints on states of affairs reduce the space of possible worlds below the full Boolean cube.

Figure 1 illustrates the relationship between the free model and constrained models.

5 Formal Limits: Expressibility and Equivalence

The deepest results of the formalization concern the boundaries of what the object language can express. These are not claims within the Tractarian system but *about* it—theorems at the meta-level that formalize Wittgenstein's saying/showing distinction.

5.1 The `expresses` Relation

We introduce a typed bridge between the object language and the metalanguage:

Definition 5.1 (`expresses`)**.** A proposition p : Proposition S *expresses* a meta-level property P : Prop if and only if p's truth value tracks P in every world:

$$\mathsf{expresses}\ p\ P \ \stackrel{\text{def}}{=}\ \forall\, w : \mathsf{World\ S},\ \ p.\mathsf{eval}\ w \leftrightarrow P$$

```
def expresses (p : Proposition S) (P : Prop) : Prop :=
  ∀ w : World S, p.eval w ↔P
```

The type signature makes the object/meta distinction visible: p lives in the object language, P lives in Lean's metalanguage, and `expresses` is the claim that they correspond. This is the formalization's key analytical device.

5.2 Expressibility Collapse

Theorem 5.2 (Expressibility collapse)**.** *If a proposition p expresses a world-independent property P : Prop, then p is a tautology or a contradiction.*

Proof. Suppose expresses p P holds. Then for any two worlds w_1, w_2:

$$p.\mathsf{eval}\ w_1 \ \leftrightarrow\ P\ \leftrightarrow\ p.\mathsf{eval}\ w_2$$

so p has the same truth value in all worlds. By excluded middle on $p.\mathsf{eval}\ w_0$ for an arbitrary world w_0, either p holds everywhere (tautology) or fails everywhere (contradiction). □

```
theorem saying_showing_triviality (q : Proposition S)
    (P : Prop) [Nonempty S]
    (h : expresses q P) :
    IsTautology q ∨IsContradiction q := by
  apply world_constant_taut_or_contra
  intro w₁ w₂
  exact (h w₁).trans (h w₂).symm
```

The contrapositive is immediate and philosophically sharper:

Corollary 5.3 (Nontrivial propositions cannot express world-independent content)**.** *If p is nontrivial (i.e., true in some world and false in another), then there is no P : Prop such that* expresses p P.

```
theorem nontrivial_cannot_express_world_independent
    (p : Proposition S) (P : Prop) [Nonempty S]
    (hnt : Nontrivial p) :
    ¬ expresses p P := by
  intro h
  exact nontrivial_expressibility_requires_world_dependence
    p P hnt h
```

This is the paper's headline result. It says: if a proposition genuinely "says" something—i.e., carves out a nontrivial region of logical space—then what it says cannot be a world-independent truth. Genuine content requires variation across possible worlds. What is world-independent can only be "shown" by the structure of the formalism itself, not "said" within it.

This transforms the saying/showing distinction from a philosophical thesis into a provable constraint on truth-functional languages.

Remark 5.4 (On the definitional character of the collapse). A natural objection is that the expressibility collapse theorem is *trivial*: since $P : \mathsf{Prop}$ is world-independent by construction (it is a closed term in Lean's metalanguage), the collapse to tautology or contradiction follows almost definitionally from the definition of `expresses`.

We acknowledge this. The proof is indeed short—two lines of tactic code. The contribution lies not in the proof's difficulty but in three aspects of the *formulation*:

(a) Making the object/meta boundary typed. The definition of `expresses` forces the distinction between $p : \mathsf{Proposition\ S}$ (object language) and $P : \mathsf{Prop}$ (metalanguage) to be a type-level constraint. This makes the boundary machine-checkable rather than merely conventional.

(b) The analogy to other definitional results in logic. Many foundational results in logic are "definitional" in a similar sense. Tarski's undefinability theorem rests on the diagonal lemma, which is itself a consequence of how truth predicates are defined. The incompleteness theorems depend on the coding of syntax. In each case, the contribution is the *conceptual apparatus* that makes the result statable, not the complexity of the derivation. Our `expresses` relation plays an analogous role: it provides the vocabulary in which the saying/showing distinction becomes a theorem.

(c) The constraint is genuinely informative. The collapse theorem rules out a precise class of claims: no nontrivial proposition can serve as a "name" for a fixed metalinguistic truth. This is not obvious from the definition of `expresses` alone—one must also establish (via `world_constant_taut_or_contra`) that world-invariant truth values force triviality.

Remark 5.5 (The `Nonempty S` assumption). Several key theorems—including the expressibility collapse (Theorem 5.2), `world_constant_taut_or_contra`, and `structEq_ne_semEq`—require `[Nonempty S]`, i.e., that the type of states of affairs is inhabited.

This assumption is philosophically motivated: a world with no possible states of affairs is degenerate in the Tractarian framework. TLP 1 ("The world is everything that is the case") presupposes that there are things that could be the case. When S is empty, the only world is the trivial predicate, every proposition is a tautology, and the interesting structure collapses. The assumption `[Nonempty S]` excludes this degenerate case.

Technically, the assumption is used to construct a witness world in which to evaluate propositions. For the collapse theorem, we need at least one world to case-split on; for the equivalence hierarchy, we need at least one atom to construct separating witnesses.

5.3 The Equivalence Hierarchy

The formalization distinguishes three notions of proposition equivalence, ordered by strictness:

Definition 5.6 (Equivalence hierarchy)**.** Let $p, q : \mathsf{Proposition\ S}$.

1. **Structural equivalence** ($p \sim_{\text{struct}} q$): p and q are built from identical constructors in the same tree shape with identical atoms.
2. **Logical-form equivalence** ($p \sim_{\text{form}} q$): there exists a bijective renaming of atoms (a permutation $\sigma : \mathsf{S} \xrightarrow{\sim} \mathsf{S}$) such that $p.\texttt{rename}\ \sigma = q$.
3. **Semantic equivalence** ($p \sim_{\text{sem}} q$): p and q have the same truth value in every world: $\forall w,\ p.\mathsf{eval}\ w \leftrightarrow q.\mathsf{eval}\ w$.

We prove that $\sim_{\text{form}}$ is an equivalence relation (reflexive by the identity permutation, symmetric by inverse, transitive by composition) and establish the strict hierarchy:

Theorem 5.7 (Strict separation)**.** $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$

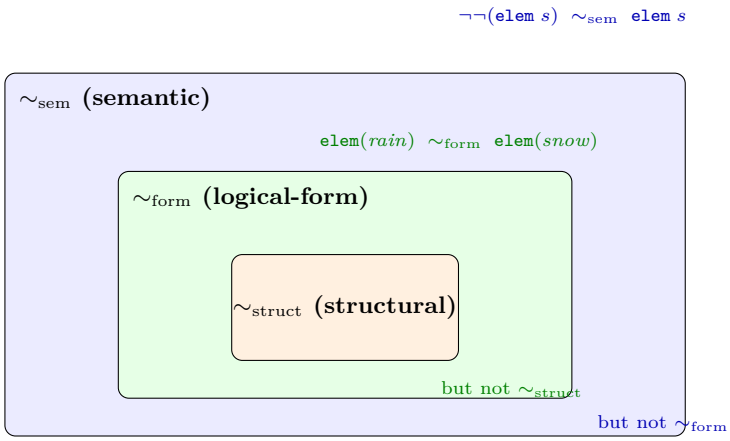

Figure 2: The equivalence hierarchy with strict separation witnesses. Each inclusion is proper: $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$.

Proof. Each inclusion is witnessed by containment and a separating example.

$\sim_{\text{struct}} \subsetneq \sim_{\text{form}}$*:* Structural equality implies propositional equality (by induction), which implies logical-form equivalence via the identity permutation. Strictness: `elementary`(*rain*) and `elementary`(*snow*) are $\sim_{\text{form}}$ (via the swap permutation) but not $\sim_{\text{struct}}$ (different atoms).

$\sim_{\text{form}} \subsetneq \sim_{\text{sem}}$*:* Logical-form equivalence implies truth-table isomorphism (Theorem `formEq_implies_truth_table_iso`): there exists a world-relabeling f such that $p.\mathsf{eval}\ w \leftrightarrow q.\mathsf{eval}\ (f\ w)$ for all w. When this relabeling is the identity, semantic equivalence follows. However, $\sim_{\text{form}}$ does *not* imply $\sim_{\text{sem}}$ in general—the relabeling need not be trivial. Strictness: $\neg\neg$(`elementary` s) is $\sim_{\text{sem}}$ to `elementary` s (by double negation) but not $\sim_{\text{form}}$ (renaming preserves tree depth, so a depth-3 tree cannot become depth-1). □

Figure 2 illustrates the hierarchy. This clarifies a long-standing ambiguity in the *Tractatus*. Wittgenstein's notion of "logical form" (TLP 4.0141) is neither syntactic identity nor truth-conditional equivalence but something in between. Our $\sim_{\text{form}}$—defined via bijective atom renaming—provides a formal candidate: it captures the connective structure while abstracting over which particular atoms appear. The strict separation proves that this intermediate notion is genuine, not collapsible into either extreme.

5.4 First-Order Extension

In a companion file (`TractatusQuantifiers.lean`), we extend the propositional calculus with first-order quantifiers using higher-order abstract syntax (HOAS):

```
inductive FOProp (S : Type) (D : Type) : Type where
  | lift    : Proposition S →FOProp S D
  | negFO   : FOProp S D →FOProp S D
  | conjFO  : FOProp S D →FOProp S D →FOProp S D
  | forall_ : (D →FOProp S D) →FOProp S D
  | exists_ : (D →FOProp S D) →FOProp S D
```

The HOAS encoding leverages Lean's metalanguage for variable binding, avoiding explicit substitution machinery. This substantially reduces proof overhead but ties the formalization more tightly to Lean's metatheory than a de Bruijn index encoding would—in particular, induction over HOAS terms requires care, since the recursor for HOAS inductives must track the functional argument.

We prove that compositionality lifts to the first-order level:

Theorem 5.8 (First-order compositionality)**.** *For any* $p : \mathsf{FOProp}\ \mathsf{S}\ D$ *and worlds* w_1, w_2 *agreeing on all states of affairs,* $p.\mathsf{eval}\ w_1 \leftrightarrow p.\mathsf{eval}\ w_2$.

We also establish eight additional results: universal–existential duality (TLP 5.52), universal distribution over conjunction, existential distribution over disjunction (reverse direction), lift preservation, vacuous quantifier elimination (requiring `Nonempty D`), double negation at the first-order level, De Morgan duality for quantifiers, and first-order bivalence.

This section is preliminary. The results proved are standard consequences of the semantic definitions. The philosophically interesting question—whether Wittgenstein's N-operator can ground quantification over infinite domains—requires a separate formalization of the N-operator, which is deferred to future work. The file notes the Geach–Soames critique [8, 9]: for infinite domains, Wittgenstein's claim that quantifiers reduce to applications of the N-operator is problematic, since the N-operator is a finitary syntactic operation. For finite domains the connection is unproblematic and will be established when the N-operator formalization is available.

6 Related Work

Formal reconstructions of the *Tractatus*. Lokhorst [3] provides the most comprehensive prior formal reconstruction, covering ontology, semantics, and propositional attitudes in a set-theoretic framework. His axiomatization of independence and his truth-functionality theorem (every sentence is a truth-function of elementary sentences) overlap with our encoding, but he does not formalize the saying/showing boundary itself. Weiss [7] gives a 50-page reconstruction of the *Tractatus*'s logical system, analyzing the N-operator and form-series in detail and proving that tautology-hood is Π^1_1-complete for countable domains—a striking complexity result that sharpens the Geach–Soames critique. Miller [10] develops a Tractarian semantics for predicate logic that is independent of the metaphysics of logical atomism, engaging with both Fogelin's [4] critique of expressive completeness and Carruthers' [5] book-length treatment of Tractarian semantics. Stokhof [6] reads the *Tractatus* as a proto-formal semantics, tracing its influence through Carnap's state descriptions to possible-worlds semantics, and argues that the saying/showing distinction arises from the universalism of the framework—a philosophical diagnosis our expressibility collapse theorem makes precise. All of these works provide formal clarity but are carried out on paper; none offers machine-checked proofs, and none parameterizes over world models to isolate which claims are structural invariants versus modeling assumptions.

The expressive completeness debate. Fogelin [4] initiates the modern debate over whether the *Tractatus*'s logical resources are expressively complete. Geach [8] argues that the N-operator cannot ground quantification over infinite domains, since it is a finitary operation. Soames [9] sharpens this: the *Tractatus*'s claim that all of logic reduces to truth-functional operations on elementary propositions is undermined when the domain of quantification is infinite. Carruthers [5] provides a sympathetic reconstruction of Tractarian semantics that attempts to address some of these concerns. Our formalization takes the critique seriously: the first-order extension (Section 5.4) uses standard quantifiers rather than the N-operator, and notes the finite-domain connection as a separate theorem to be proved only when an N-operator formalization is available.

Proof assistants and philosophy. Benzmüller and colleagues [11] have used Isabelle to formalize ontological arguments and modal logic. Their approach uses shallow semantic embeddings of higher-order modal logic into Isabelle's metalogic, which contrasts with our direct encoding in dependent type theory: we define the Tractarian object language explicitly as an inductive type, whereas shallow embeddings reuse the proof assistant's logic as the object logic. Fitelson and Zalta [12] pioneer "computational metaphysics," using automated reasoners to verify theorems in abstract-object theory. Scherf [13] provides the most directly comparable work: a machine-checked axiomatization of Advaita Vedānta in Lean 4 (69 axioms, 40+ theorems), demonstrating that complete philosophical systems can be formalized in dependent type theory. Corfield [14] makes a sustained case for homotopy type theory as a foundation for philosophical reasoning, complementing the formal-verification approach taken here. To our knowledge, no prior work formalizes any part of the *Tractatus* in any proof assistant, nor uses the type system to isolate the saying/showing boundary as a provable constraint.

7 Discussion

7.1 What the Formalization Shows

The formalization yields three kinds of insight.

First, it makes the *Tractatus*'s implicit modeling decisions explicit. The choice $\mathsf{World}\ \mathsf{S} = \mathsf{S} \to \mathsf{Prop}$ is not neutral: it builds independence into the foundations. By parameterizing over `WorldModel`, we separate what follows from the propositional syntax (compositionality) from what follows from the choice of world space (independence). This distinction is available in principle but rarely made precise in the philosophical literature.

Second, it transforms the saying/showing distinction from a metaphilosophical thesis into a theorem with a two-line proof. The `expresses` relation is the key: once the object/meta boundary is a type distinction, the collapse result (Theorem 5.2) follows by transitivity of biconditional. See Remark 5.4 for a discussion of the definitional character of this result and why the formulation is nonetheless a genuine contribution.

Third, the equivalence hierarchy (Theorem 5.7) provides a precise formal candidate for Wittgenstein's notion of "logical form." The strict separation $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$ shows that logical form, understood as connective structure up to atom permutation, is a genuine intermediate notion that cannot be reduced to either syntax or semantics.

7.2 Limitations

Several aspects of the *Tractatus* lie beyond the scope of this formalization.

Object structure. The type S is abstract: we do not model how objects combine into states of affairs. TLP 2.0141 ("The possibility of its occurrence in atomic facts is the form of the object") suggests a richer dependent-type encoding where the form of an object constrains its combinatorial possibilities. This would make independence a substantive theorem rather than a definitional consequence.

The N-operator. Wittgenstein's claim that all propositions can be generated from elementary propositions by successive applications of a single operation (TLP 6) is not formalized here. The N-operator is a finitary syntactic construction, and its connection to quantification over infinite domains remains philosophically contentious (Section 5.4).

The picture theory. The *Tractatus*'s claim that propositions are *pictures* of reality (TLP 2.1–2.174) involves a notion of structural isomorphism between language and world that goes beyond truth-functional evaluation. Our formalization captures the truth-conditional aspect but not the pictorial-isomorphism aspect.

Self-reference and the ladder. TLP 6.54 famously declares that the propositions of the *Tractatus* itself are "nonsensical" and must be "thrown away" after use. Our formalization is, by design, a metalanguage that talks *about* the Tractarian object language. It cannot formalize its own status as a ladder to be discarded—this is the point where type theory, like any formal system, reaches its own saying/showing boundary.

We mark this boundary with a deliberate axiom:

```
axiom silence : True
```

This is not a missing proof but a design boundary. The axiom states nothing contentful (`True` is already provable); it marks the point where the formalization acknowledges its own limits.

Remark 7.1 (Why `True`). The choice of `True` as the axiom's type is deliberate. Because `True` is already provable in Lean (via `trivial`), the axiom `silence` adds no logical content whatsoever—it introduces no new assumptions and changes no proof obligations. Its content is its *label*, not its type. The name "silence" marks a boundary; the type `True` ensures that boundary is inert. This mirrors TLP 7's status as a proposition that, by Wittgenstein's own account, says nothing: it is a gesture toward what lies beyond the expressible, not an assertion within it.

7.3 Beyond the Tractatus

The expressibility collapse theorem (Theorem 5.2) is formulated in terms of Tractarian propositions, but it depends only on two structural features: (i) the object language is truth-functional (proposition truth values are determined by evaluation against worlds), and (ii) the metalanguage is typed, so that a "world-independent property" is simply a term of type Prop with no free world variable. Any truth-functional object language embedded in a typed metalanguage with this structure will exhibit the same collapse.

This means the result is not specific to the *Tractatus*. Classical propositional logic, Boolean algebras interpreted over sets of valuations, and modal propositional logics (when the accessibility relation is removed) all satisfy the preconditions. The collapse theorem is a general fact about the typed embedding of truth-functional languages: the type system enforces a gap between world-dependent content (nontrivial propositions) and world-independent truth (metalinguistic facts), and no nontrivial proposition can bridge it.

For logicians who may not share an interest in Wittgenstein exegesis, this generality is the main takeaway: the saying/showing distinction, stripped of its philosophical mystique, is a typing constraint that any truth-functional formalization must respect.

7.4 Future Work

Several directions for further investigation arise naturally. First, formalizing the N-operator and proving the finite-domain quantifier connection (that $\forall x.\, P(x)$ reduces to an N-application over a `Fintype` domain) would complete the truth-functional picture for the finite case and sharpen the boundary with the infinite case. Second, encoding object structure via dependent types—so that states of affairs are typed combinations of objects constrained by form (TLP 2.0141)—would make independence a substantive theorem rather than a definitional consequence of the world type. Third, extending the `expresses` relation to world-dependent properties ($P : \mathsf{World\ S} \to \mathsf{Prop}$) would allow a finer analysis of what nontrivial propositions *can* express, complementing the current result about what they cannot. Fourth, formalizing the doxastic extension of Tractarian semantics along the lines of Lokhorst [3] would test whether the invariant/assumption/limit decomposition extends to propositional attitudes.

8 Conclusion

We have presented a machine-checked reconstruction of the semantic core of *Tractatus Logico-Philosophicus* in Lean 4. The formalization proves 50 formally verified results across two files, decomposes the *Tractatus* into structural invariants, model-dependent assumptions, and formal limits of expressibility, and characterizes the saying/showing distinction as a theorem: nontrivial propositions cannot express world-independent truths.

The equivalence hierarchy $\sim_{\text{struct}} \subsetneq \sim_{\text{form}} \subsetneq \sim_{\text{sem}}$ provides a typed decomposition of the space between syntax and semantics, isolating logical form as a genuine intermediate structure.

These results suggest that proof assistants are a productive tool for philosophical analysis—not merely for verifying known results, but for revealing structural constraints that are invisible when the analysis is conducted on paper.

The formalization is publicly available at `https://github.com/rjwalters/tractatus`, with a browsable proof gallery at `https://leangenius.org/proof/tractatus-ontology`.

A Build Instructions

The formalization is built with Lean 4 using the `lake` build system. To reproduce:

1. Install Lean 4 via `elan`:
   ```
   curl -sSf https://raw.githubusercontent.com/leanprover/elan/master/elan-init.sh | sh
   ```
2. Clone the repository:
   ```
   git clone https://github.com/rjwalters/tractatus.git
   ```
3. The `lean-toolchain` file pins the version to `leanprover/lean4:v4.17.0`.

4. The `lakefile.lean` declares a dependency on Mathlib (the formalization uses `import Mathlib.Tactic`).
5. Build: `lake build`

The build produces zero errors, zero warnings, and zero `sorry` obligations. The sole axiom (`axiom silence : True`) is intentional and discussed in Section 7.2.

Proof files:

- `proofs/TractatusOntology.lean` — 1,229 lines, 41 declarations
- `proofs/TractatusQuantifiers.lean` — 288 lines, 9 declarations

B Complete Theorem Listing

Table 1 classifies every theorem and lemma declaration in the formalization. The "Class" column indicates whether the result is a structural **I**nvariant, a model-dependent **A**ssumption or counterexample, or a formal **L**imit on expressibility. Results marked **H** are helper lemmas for the equivalence hierarchy. Results marked **E** are engineering artifacts (Bool-valued evaluation). All Lean names are given in full; they can be located in the source files by exact text search.

References

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Table 1: Complete listing of all 50 theorem/lemma declarations plus 1 axiom, classified by the three-way decomposition. **I** = invariant, **A** = assumption/counterexample, **L** = limit, **H** = hierarchy helper, **E** = engineering.

#	Lean name	TLP	File	Cl.
TractatusOntology.lean (41 declarations)				
1	`evalBool_correct`	5.1	Ont.	E
2	`truth_functional_compositionality_gen`	5	Ont.	I
3	`evalM_free_eq_eval`	—	Ont.	I
4	`tautology_is_world_invariant`	6.1	Ont.	I
5	`contradiction_holds_nowhere`	4.46	Ont.	I
6	`elem_prop_bivalence`	4.023	Ont.	I
7	`truth_functional_compositionality`	5	Ont.	I
8	`elementary_independence`	2.061	Ont.	A
9	`double_negation`	5.101	Ont.	I
10	`de_morgan_disj`	5.101	Ont.	I
11	`de_morgan_conj`	5.101	Ont.	I
12	`excluded_middle_tautology`	4.46	Ont.	I
13	`conj_neg_contradiction`	4.46	Ont.	I
14	`impl_semantics`	5.101	Ont.	I
15	`biimp_semantics`	5.101	Ont.	I
16	`modus_ponens`	4.121	Ont.	I
17	`nand_expresses_neg`	5.5	Ont.	I
18	`nand_expresses_conj`	5.5	Ont.	I
19	`constrained_independence_fails`	2.061	Ont.	A
20	`weather_independence_fails`	2.061	Ont.	A
21	`rename_id`	4.014	Ont.	H
22	`rename_comp`	4.014	Ont.	H
23	`rename_eval`	4.014	Ont.	H
24	`formEq_refl`	4.0141	Ont.	H
25	`formEq_symm`	4.0141	Ont.	H
26	`formEq_trans`	4.0141	Ont.	H
27	`eq_of_structEq` (private)	4.0141	Ont.	H
28	`structEq_implies_formEq`	4.0141	Ont.	L
29	`formEq_implies_truth_table_iso`	4.0141	Ont.	L
30	`truth_table_iso_id_implies_semEq`	4.0141	Ont.	L
31	`formEq_not_structEq_witness`	4.0141	Ont.	L
32	`semEq_not_formEq_witness`	4.0141	Ont.	L
33	`structEq_ne_semEq`	4.0141	Ont.	L
34	`world_constant_taut_or_contra`	4.46	Ont.	L
35	`saying_showing_triviality`	4.12	Ont.	L
36	`contingent_propositions_vary`	4.46	Ont.	L
37	`nontrivial_of_not_taut_not_contra`	4.46	Ont.	L
38	`nontrivial_iff_not_taut_not_contra`	4.46	Ont.	L
39	`proposition_seven`	7	Ont.	L
40	`nontrivial_expressibility_requires_world_dependence`	4.12	Ont.	L
41	`nontrivial_cannot_express_world_independent`	4.12	Ont.	L
TractatusQuantifiers.lean (9 declarations)				
42	`truth_functional_compositionality_fo`	5	Quant.	I
43	`forall_exists_duality`	5.52	Quant.	I
44	`forall_distrib_conj`	5.52	Quant.	I
45	`exists_distrib_disj_reverse`	5.52	Quant.	I
46	`lift_eval`	—	Quant.	I
47	`forall_vacuous`	—	Quant.	I
48	`double_negation_fo`	5.101	Quant.	I
49	`not_exists_forall_not`	5.52	Quant.	I
50	`fo_bivalence`	4.023	Quant.	I
Axiom				
—	`axiom silence : True`	7	Ont.	L

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